

UNIT – II LONG ANSWERS

1. Explain First-Fit, Best-Fit, and Worst-Fit Memory Allocation with Example

Memory allocation is the process of assigning memory blocks to processes.

Example

Available Memory Blocks:

Block Size (KB)

B1 100

B2 500

B3 200

B4 300

B5 600

Processes:

Process Size (KB)

P1 212

P2 417

P3 112

P4 426

First-Fit:

Allocates the first block that is large enough.

P1 → B2 (500)

P2 → B5 (600)

P3 → B2 remaining space

P4 → Not allocated

Advantages

Fast allocation.

Simple implementation.

Disadvantages

Causes external fragmentation.

Best-Fit:

Allocates the smallest suitable block.

P1 → B4 (300)

P2 → B2 (500)

P3 → B3 (200)

P4 → B5 (600)

Advantages:

Reduces wasted memory.

Disadvantages:

Requires searching entire memory.

Worst-Fit:

Allocates the largest available block.

P1 → B5 (600)

P2 → Remaining part of B5

P3 → B2 (500)

Advantages

Leaves larger free blocks.

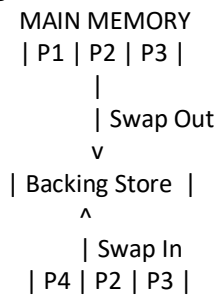
Disadvantages

Poor memory utilization.

2. Explain Swapping with Suitable Diagram

Definition: Swapping is a memory management technique in which processes are temporarily moved from main memory to secondary storage and brought back when required.

Diagram:



Advantages:

Increases degree of multiprogramming.

Improves CPU utilization.

Disadvantages

High disk I/O overhead.

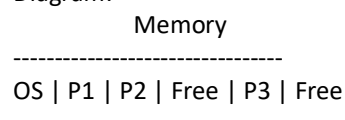
Slower execution.

3. Explain Contiguous Memory Allocation

Definition:

Each process occupies a single contiguous block of memory.

Diagram:



Types:

Fixed Partitioning

Memory divided into fixed-size partitions.

Dynamic Partitioning

Partitions created according to process size.

Advantages

Simple implementation.

Fast access.

Disadvantages

External fragmentation.

Difficult process expansion.

4. Explain Paging Hardware

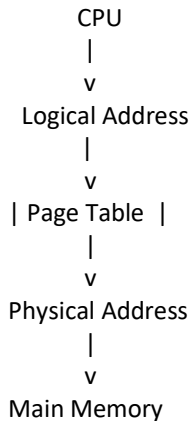
Definition: Paging is a memory management technique that divides memory into fixed-size pages and frames.

Hardware Components:

- CPU
- Page Table

- Memory Management Unit (MMU)
- Main Memory

Diagram:



Advantages:

Eliminates external fragmentation.

Supports virtual memory.

5. Explain Paging Memory Management with Example

Definition:

Paging divides logical memory into pages and physical memory into frames.

Example:

Page Size = 1 KB

Logical Address Space = 4 Pages

Pages

P0

P1

P2

P3

Page Table

Page Frame

0 5

1 2

2 8

3 4

Logical Address = (Page No, Offset)

Example:

(2,100)

Frame = 8

Physical Address = (8,100)

Advantages

No external fragmentation.

Efficient memory utilization.

6. Explain Segmentation with Neat Diagram

Definition: Segmentation divides a program into logical units called segments.

Diagram: Reffer class notes

Segment Table

Segment Base Limit

Segment Base Limit

Code 1000 400

Data 2000 300

Stack 3000 200

Advantages:

Logical view of memory.

Supports protection and sharing.

Disadvantages

External fragmentation.

7. Explain Segmentation Hardware

8. Write a Short Note on Fragmentation

Definition:

Fragmentation occurs when memory becomes divided into unusable pieces.

Types:

- External Fragmentation

Free memory exists but is scattered.

Free P1 Free P2 Free

Solution

Compaction

- Internal Fragmentation

Allocated block is larger than required.

Example:

Process = 18 KB

Allocated Partition = 20 KB

Unused = 2 KB

Solution

Use paging.

9. Explain Segmentation with Paging

Definition:

Combines segmentation and paging.

Working:

Program divided into segments.

Each segment divided into pages.

Diagram:

Program

|

Segments

|

Pages

|

Frames

Advantages:

Eliminates external fragmentation.

Provides logical memory organization.

Disadvantages

Complex implementation.

10. Write a Short Note on Demand Paging

Definition:

Pages are loaded into memory only when required.

Working:

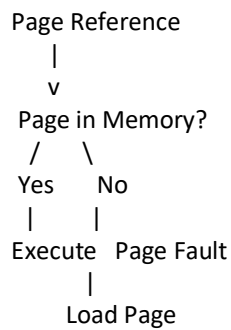
Process starts execution.

Required page not found.

Page fault occurs.

OS loads page from disk.

Diagram:



Advantages:

Reduced memory usage.

Faster program startup.

Disadvantages

Page fault overhead.

11. Write a Short Note on Disk Management

Disk management handles storage devices and free space.

Functions:(expalin)

Disk Formatting

Boot Block Management

Bad Block Recovery

Disk Scheduling

Free Space Management

Objectives:

Efficient storage utilization.

Fast access to data.

12. Explain Dynamic Relocation using Relocation Register

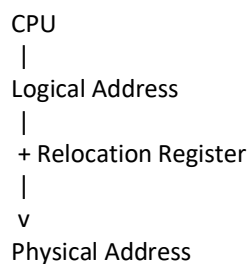
Definition:

Dynamic relocation allows programs to be loaded anywhere in memory.

Relocation Register:

Stores starting physical address.

Diagram:



Example:

Logical Address = 500

Relocation Register = 1000

Physical Address

= 1000 + 500

= 1500

Advantages:

Supports memory protection.

Flexible allocation.

13. Explain FIFO Page Replacement Policy

Definition:

FIFO (First-In First-Out) replaces the oldest page in memory. (Explain)

Example:

Reference String:

1 2 3 4 1 2 5

Frames = 3

1

1 2

1 2 3

4 2 3

4 1 3

4 1 2

5 1 2

Advantages:

Simple implementation.

Disadvantages

May suffer from Belady's Anomaly.

14. Explain LRU Page Replacement Algorithm with Example

Definition:

LRU (Least Recently Used) replaces the page that has not been used for the longest time.(Explain)

Example:

Reference String

1 2 3 4 1 2 5

Frames = 3

1

1 2

1 2 3

4 2 3

4 1 3

4 1 2

5 1 2

Working:

When a replacement is needed:

Find the least recently used page.

Replace it with the new page.

Advantages:

Better performance than FIFO.

Closer to Optimal algorithm.

Disadvantages:

Requires tracking page usage history.

15. Page Replacement Problem

Reference String:

7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1

Frames = 3

Results

Algorithm Page Faults

FIFO 15

LRU 12

Optimal 9

Conclusion:

FIFO produces maximum page faults.

Optimal produces minimum page faults.

LRU performs better than FIFO.

16. Explain Optimal Page Replacement Algorithm

Definition:

Optimal Page Replacement replaces the page that will not be used for the longest time in future.(Explain)

Algorithm:

Check future references.

Replace the page whose next use is farthest away.

If a page is never used again, replace it.

Example:

Reference String:7 0 1 2 0 3 0

Frames = 3

7

7 0

7 0 1

2 0 1

2 0 3

Advantages:

Minimum page faults.

Used as benchmark.

Disadvantages:

Future references are unknown.

Cannot be implemented practically.

17. Write a Note on Allocation of Frames

Definition:Frame allocation determines how many frames each process receives.

Types:

Equal Allocation

Each process receives equal number of frames.

Example:

Frames = 30

Processes = 3

Each gets:

[

30/3=10

]

frames.

Proportional Allocation:

Frames allocated according to process size.

$$\left[\begin{array}{l} \text{Frames}_i = \frac{\text{Size}_i}{\text{Total Size}} \times \text{Total Frames} \end{array} \right]$$

Priority Allocation:

High-priority processes receive more frames.

Advantages:

Better memory utilization.

Reduced page faults.

18. Write a Note on Thrashing

Definition

Thrashing occurs when the system spends more time handling page faults than executing processes.

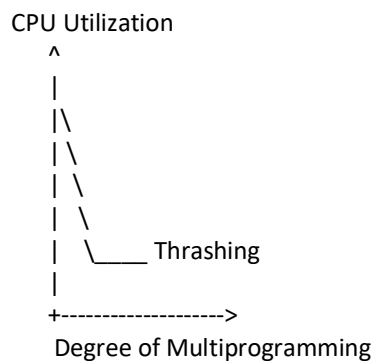
Causes

Insufficient frames.

Excessive multiprogramming.

High page-fault rate.

Diagram:



Effects:

Low CPU utilization.

Poor system performance.

Solutions:

Working Set Model.

Page Fault Frequency.

Increase memory.

19. Write a Note on Working Set Model

Definition:

Working Set is the set of pages currently required by a process.

Working Set Window:

$$\left[\begin{array}{l} \text{WSS}_i \end{array} \right]$$

represents pages referenced during last Δ references.

Example:

Reference String

1 2 3 4 2 1 5

Working Set

{1,2,3,4,5}

Advantages:

Prevents thrashing.

Improves performance.

20. FIFO Steps Showing Page Faults

Reference String:

7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1

Frames = 3

Ref Frames Fault

7	7	-
0	7 0	-
1	7 0 1	✓
2	2 0 1	✓
0	2 0 1	X
3	2 3 1	✓
0	2 3 0	✓
4	4 3 0	✓
2	4 2 0	✓
3	4 2 3	✓
0	0 2 3	✓
3	0 2 3	X
2	0 2 3	X
1	0 1 3	✓
2	0 1 2	✓
0	0 1 2	X
1	0 1 2	X
7	7 1 2	✓
0	7 0 2	✓
1	7 0 1	✓

Total FIFO Page Faults:

21. Page Replacement Problem

Reference String

7,2,3,1,2,5,3,4,6,7,7,1,0,5,4,6,2,3,0,1

Frames = 3

Results

Algorithm Page Faults

FIFO 17

LRU 18

Optimal 11

22. Page Replacement Problem

Reference String

1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6

Frames = 4

Results

Algorithm Page Faults

Algorithm Page Faults

FIFO 10

LRU 8

Optimal 7

23. Explain FCFS Disk Scheduling with Example

Definition:

First Come First Serve services requests in arrival order.

Example

Queue:

98,183,37,122,14,124,65,67

Head = 53

Movement

53→98→183→37→122→14→124→65→67

Advantages

Simple.

Fair.

Disadvantages

High seek time.

24. Explain SSTF Disk Scheduling with Example

Definition

Shortest Seek Time First serves nearest request first.

Example

Head = 53

Queue:

98,183,37,122,14,124,65,67

Order:

53→65→67→37→14→98→122→124→183

Advantages:

Reduced seek time.

Disadvantages

Starvation possible.

25. Explain SCAN Disk Scheduling

Definition

Disk arm moves in one direction servicing requests until end track then reverses.

Diagram

0 ----- 199

^

Head

Movement:

53→65→67→98→122→124→183→199

↓

reverse

Advantages:

Uniform waiting time.

26. Explain C-SCAN Disk Scheduling

Definition

Circular SCAN services requests in one direction only.

Working

0 ----- 199-->

Reach end
Jump to start
Continue

Advantages:
More uniform waiting time.

Disadvantages:
Additional jump overhead.

27. Explain LOOK Disk Scheduling

Definition

LOOK is similar to SCAN but reverses at the last request instead of disk end.

Example

53→65→67→98→122→124→183

Reverse from 183 directly.

Advantages

Less seek time than SCAN.

28. Explain C-LOOK Disk Scheduling

Definition

Circular LOOK services requests in one direction and jumps to lowest request.

Example

53→65→67→98→122→124→183

Jump

14→37

Advantages:

Better than C-SCAN.

Reduced unnecessary movement.

29. Disk Scheduling Numerical

Queue:

55,58,39,18,90,160,150,38,184

Initial Head = 100

Total Head Movements

Algorithm Movement

FIFO 498

SSTF 248

SCAN 280

C-SCAN 388

Average Seek Length

(9 requests)

Algorithm Average Seek Length

FIFO 55.33

SSTF 27.56

SCAN 31.11

C-SCAN 43.11

30. Disk Scheduling Numerical

Queue:

98,183,37,122,14,124,65,67

Initial Head = 55

Total Head Movement

Algorithm Movement

FIFO 640

SSTF 236

LOOK 299

C-LOOK 322

Average Seek Length

(8 requests)

Algorithm Average Seek Length

FIFO 80.00

SSTF 29.50

LOOK 37.38

C-LOOK 40.25

Conclusion

SSTF gives minimum seek time.

FIFO gives maximum seek time.

LOOK performs better than SCAN.

C-LOOK provides more uniform waiting time.